YASHAS CHANDRA BATHINI *Software Engineer | AI Engineer | Full-Stack Development | LLM, RAG & Cloud Systems*

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yashas144.github.io/web F-1 OPT Eligible (June 2026)

PROFILE

Software Engineer and AI Engineer with 4+ years of experience developing scalable full-stack applications, AI-powered systems, and cloud-native platforms. Skilled in Python, FastAPI, React, machine learning, Generative AI, LLM/RAG systems, and Azure. Proven ability to design, build, and deploy production-grade solutions spanning intelligent automation, predictive analytics, recommendation systems, real-time data pipelines, and modern web applications.

SKILLS

Generative AI & LLMs: Transformers, Hugging Face, RAG, Agentic AI, LangChain, NLP, FAISS, Chroma, Pinecone.

Full-Stack Web Development: React, FastAPI, REST APIs, LLM Integration, scalable API development.

Programming: Python, Java, PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn.

Data Engineering: Data Analysis, Data Visualization, Tableau, MySQL, Snowflake.

MLOps: MLflow, Time-Series Forecasting (LSTM), Anomaly Detection (Isolation Forest), XGBoost, Regression, Clustering.

Cloud & DevOps: Azure, AWS, CI/CD Pipelines, GitHub Actions, Docker, Git, Jira.

WORK EXPERIENCE

AI Engineer 06/2022 – Present
Sysarket Hyderabad, India

Key Contributions

- Designed and delivered AI-powered applications spanning predictive maintenance, recommendation systems, generative AI research, and sports analytics.
- Built scalable full-stack applications using React, FastAPI, Python, Azure, and modern AI frameworks.
- Developed production-grade APIs, retrieval systems, machine learning pipelines, and cloud-native deployments supporting real-world business use cases.
- Collaborated across the full software lifecycle including architecture, development, deployment, monitoring, and optimization.

Auravance - The Future of Industrial Foresight

- Building a full-stack AI platform (React, FastAPI) for real-time equipment monitoring and failure prediction using XGBoost and LSTM, improving detection accuracy by 30%.
- Developing a RAG-based root cause analysis system over equipment manuals, generating explainable diagnostics.
- Designing RUL (Remaining Useful Life) prediction and anomaly detection (Isolation Forest) to reduce downtime risk.
- Engineering scalable real-time IoT data pipelines using MQTT and Python to ingest, process, and store high-frequency sensor telemetry, enabling anomaly detection, predictive forecasting, and operational insights.

AI-Powered Music Intelligence Platform | Live Demo

- Designed and delivered an AI-powered music recommendation platform leveraging Hybrid-RAG (BM25 + vector search) across 500K+ songs, enabling context-aware discovery based on mood, genre, and user intent.
- Architected a scalable RAG pipeline with embedding-based retrieval and ranking, significantly improving recommendation relevance and reducing noise in outputs.
- Engineered real-time features, including real-time mood detection and seamless playback.
- Deployed a cloud-native full-stack system (React + FastAPI) on Azure and Hugging Face with CI/CD pipelines, ensuring reliable and scalable delivery.

Deep Learning-Based Image Synthesis with GAN Architecture | Research Publication

- Developed GAN-based image synthesis systems using PyTorch and TensorFlow, improving output quality through architecture tuning and training optimization.
- Contributed to applied generative AI research, developing GAN architectures for real-world applications and publication-ready experimentation.

AI-Based Sports Analytics Platform

- Developed a robust sports analytics platform featuring ML-based match predictions, APIs, and dashboards.
- Improved model accuracy by 18% using feature engineering, hyperparameter tuning, and MLflow tracking.
- Implemented model registry, versioning, and monitoring (performance, confidence, and feature drift).
- Deployed a containerized full-stack application using React (frontend) and Flask (backend) with integrated ML services, leveraging Docker and Azure with Azure SQL DB for scalable, production-ready model serving.

EDUCATION

Master of Science - Data Science 08/2024 – 05/2026
University of North Texas, GPA: 4.0 / 4.0 Denton, Texas

Bachelor of Technology - Information Technology 08/2020 – 05/2024
Jawaharlal Nehru Technological University, Hyderabad, CGPA: 3.7 / 4 Hyderabad, India

CERTIFICATIONS

Generative AI Fundamentals - Databricks | Machine Learning with Big Data - UC San Diego | Introduction to Data Science - Cisco | Career Essentials in Generative AI - Microsoft | Python Essentials - Cisco